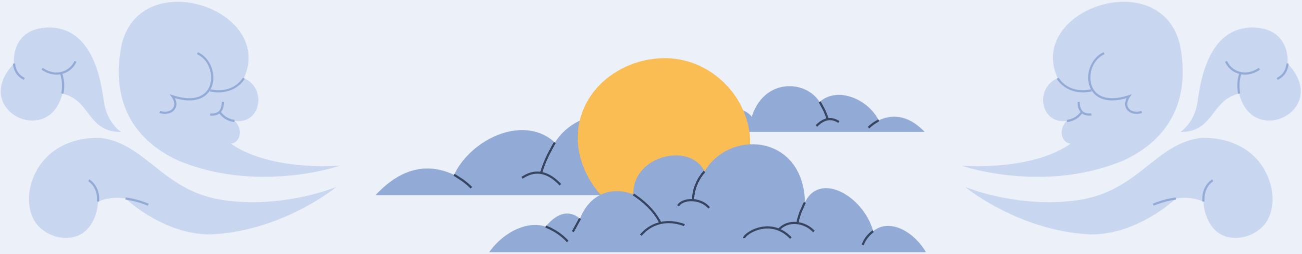


Evaluating Weather Forecasting Models During Extreme Weather Events

Jenny Xu, Tugrul Karabulut

01

Introduction

Climate Change & Weather Forecasting

- Weather forecasting is a crucial part of human life; it influences everything from daily life to agricultural planning
- Global climate change has led to increases in the frequency and severity of extreme weather events
    - Sequential extreme weather events have cascading effects that compound their consequences even further than those of just isolated events
- More important than ever to develop timely weather forecasts that can accurately predict these events in order for us to respond
- As these events are exacerbated by climate change beyond our natural climate variability, they become even harder to predict

02

Current Climate Models & Limitations

Traditional Numerical Prediction Models

- Rely on **complex numerical simulations** of atmospheric physics
  - **Resolution limitations:** Not **computationally feasible** to accurately capture very localized events like the precise evolution of a hurricane
    - Less practically useful for extreme weather events

Traditional Deep Learning Approaches

- Deterministic models have achieved **SOTA performance** in medium-range global weather prediction while having **computational costs that are several magnitudes lower**
- However, they often act as **data-driven black-box models** that **neglect the underlying physics**

ClimODE

- A spatiotemporal continuous-time PDE climate and weather model
- Models precise weather evolution with **physics-conserving** dynamics
- Uses a **computationally practical** method to solve the continuity equation for weather over the entire Earth as a system of neural ODEs
- **Outperforms** existing data-driven methods in forecasting with an order of magnitude smaller parameterization, establishing a new SOTA

Extreme Weather Prediction

Despite advances in ML models, predicting extreme weather events is still a critical and challenging issue in weather forecasting

- AI methods result in smoother forecasts, which risk underestimating the magnitude of extreme weather events
- Popular existing weather benchmarks are typically focused on general weather prediction
    - Extreme climate events are driven by complex interactions that are not seen in these simplified and controlled benchmarks
- ML models struggle to generalize to extreme weather events due to their limited representation in available data
- Current SOTA ML models either lack testing in extreme weather or have only been tested on a limited range of extreme weather events
- Remains uncertain whether models can reliably forecast weather patterns amidst the profound and unpredictable climate changes anticipated in the coming decades

How can the performance of ML models on extreme weather events be improved?

Steps

How do deterministic SOTA ML models generalize to extreme weather events?

Can fine-tuning with extreme weather data improve generalization?

03

Methods

Model

Use **ClimODE** as our base model and evaluate its performance in predicting extreme weather events

- Physics-guided AI approaches can increase model consistency with the physical world, especially on unseen or extreme events.
- More computationally efficient than other SOTA models and can be trained on a single GPU

Methods Overview

Data

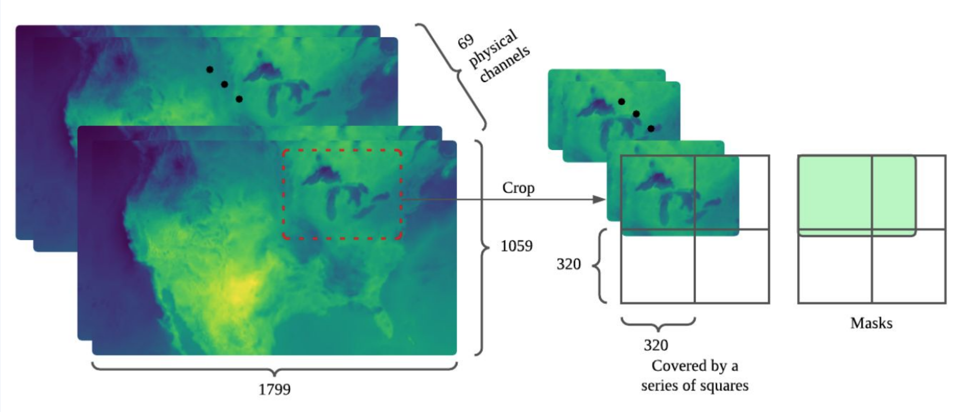
- Data collection from HRRR (High-Resolution Rapid Refresh)
- HR-Extreme preprocessing

Model

- Train and test ClimODE on high-resolution weather data from HRRR
- Evaluate ClimODE on HR-Extreme for extreme-weather performance evaluation
- Fine-tune ClimODE on a subset of HR-Extreme
- Evaluate fine-tuned ClimODE on a subset of HR-Extreme

HR-Extreme Benchmark

- New weather dataset specifically focused on extreme weather
- High-resolution: 320x320 crops from 1799 × 1059 grid (Each is pixel is 3kmx 3km)
- (As opposed ERA5: 32X64 global grids)
- 3-hour window of data: 2-hour prior to event and event's hour
- 17 extreme weather types extracted from various reports and databases
-



Training Data Collection from HRRR

High-Resolution Rapid Refresh

- Original data source that HR-Extreme was derived from
- Used to retrain ClimODE as ClimODE's regional model weights are not available
- Hourly high-resolution US dataset
- Data extraction:
    - Select a random 320x320 crop and random start and end time
    - Retrieve the 3-hour windows.

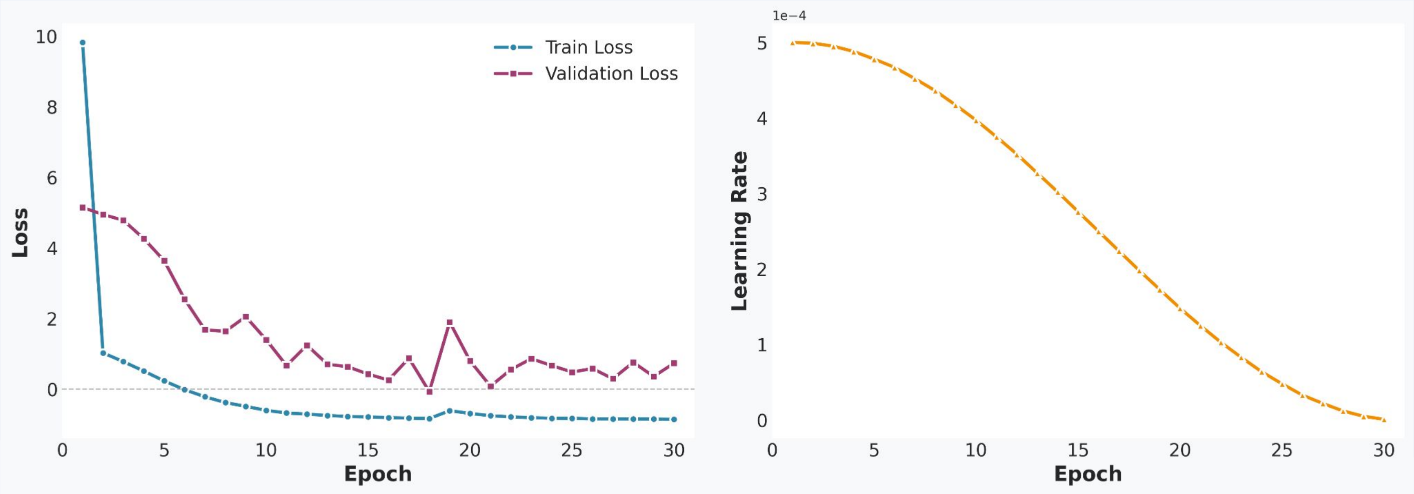
Datasets Overview

- Training: HRRR (80% train, 10% validation, 10% test)
- Testing and fine-tuning: HR-Extreme (20% train, 10% validation, 70% test)

	HRRR	HR-Extreme
Crops	8,372	16,302
Period	Jan-Jun 2020	Jul-Dec 2020
Size	31 GB	48 GB
Interval	1 h	1 h
Window size	3	3
Resolution (px)	320x320	320x320
Resolution (km)	3x3 km	3x3 km

Model Training

- Trained ClimODE on 3 NVIDIA A40 48GB GPUs for 30 epochs
- Used a batch size of 4 and learning rate of 0.0005 with an LR scheduler
- Saved the model checkpoint from epoch with the best validation error



04

Experiment & Results

Performance: HRRR vs. HR-Extreme

- RMSE: Root-Mean Squared Error
  - Model performance in capturing spatial or climate patterns
- ACC: Anomaly Correlation Coefficient
  - Model's ability to predict deviations from normal conditions
- **t2m:** Surface temperature, **u10, v10:** 10-meter wind components
- **z:** Geopotential, **t:** Atmospheric temperature

Dataset	**t2m**		**u10**		**v10**		**z**		**t**	
	RMSE	ACC	RMSE	ACC	RMSE	ACC	RMSE	ACC	RMSE	ACC
HRRR	17.01	0.43	11.79	0.22	8.66	0.26	130.98	0.46	12.66	0.50
HR-Extreme	15.19	0.35	11.79	0.26	9.05	0.30	167.56	0.43	15.39	0.45

Fine-tuning: Performance on HR-Extreme

- **t2m:** Surface temperature, **u10, v10:** 10-meter wind components
- **z:** Geopotential, **t:** Atmospheric temperature

Dataset	t2m		u10		v10		z		t	
	RMSE	ACC	RMSE	ACC	RMSE	ACC	RMSE	ACC	RMSE	ACC
HRRR	17.01	0.43	11.79	0.22	8.66	0.26	130.98	0.46	12.66	0.50
HR-Extreme	15.19	0.35	11.79	0.26	9.05	0.30	167.56	0.43	15.39	0.45
HR-Extreme (Fine-tuned)	13.28	0.25	7.90	0.40	6.46	0.43	122.39	0.37	9.98	0.47

05

Conclusions

Conclusion

- Investigated the performance of ClimODE under extreme weather conditions
    - Trained ClimODE on high-resolution weather data from HRRR
    - Compared the performance on HRRR and HR-Extreme datasets
        - Performance on both datasets was similar but not competitive due to the small training set covering a short period of time
- Fine-tuned ClimODE on HR-Extreme benchmark
    - Showed improvements over the pre-trained model on HR-Extreme and HRRR

    ⇒ can leverage ClimODE and fine tuning to improve extreme-weather performance

Future Work

- Comparison with other ML models (GNN and transformer) to evaluate if physics-informed model has an advantage in predicting extreme weather
- Could be greatly improved by training the model with long-term data using grids that cover the entire US
- Geometric/graph models for better handling of vector quantities and structure

Questions?